No of Pages : 3 **Course Code : 23MX15**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, MARCH 2025

MCA Semester : 1

23MX15 WEB TECHNOLOGIES

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.

1. a) State the purpose of Cookies in web applications.

   b) i) How does User Datagram Protocol (UDP) differ from TCP, and in what scenarios is UDP preferred over TCP? (4)

   ii) What is the layered architecture of the TCP/IP model, and how do the layers interact to enable data communication across networks? (6)

   c) i) Discuss the purpose of Idempotent HTTP methods GET, PUT and Delete with an example code. Also explain the purpose of HTTP cache and its types.

   (OR)

   ii) Discuss the roles and functionalities of web crawlers, web cookies, and search engines in the modern web ecosystem. Discuss their impact on user experience, privacy, and data management. Provide relevant examples to support your answer.

2. a) How do you create a hyperlink in HTML? What does the target="_blank" attribute do in an anchor tag?

   b) i) Why the viewport is important in web design. Give an example. (3)

   ii) Discuss the differences between inline styles, internal style sheets, and external style sheets in web development. Additionally, design a following table with appropriate styles, such as cell padding, background colors, fonts, and borders for the given data, using an external CSS file with a background image named 'image.jpg'." (7)

Roll No	Name	Team
1001	John	Red
1002	Peter	Blue
1003	Henry	Green
1004	Ford	Yellow

c) Write a complete HTML & CSS code to generate the hotel reservation form. The form fields must be validated using pattern attribute of the HTML. Provide the tool tip or error message to the user to enter the correct data. If all the fields are valid, click of Submit button should display the registration success message in the next web page.

3. a) Determine the output of the following code snippet and justify the reason for it.

```
const greet = (name = "Guest", greeting = "Hello")
=> {
   console.log(`${greeting}, ${name}!`);
    };

greet();
greet("Mohan");
greet("Mohan", "Welcome");
```

b) i) Trace the following code and determine error or output in each line. (4)

```
let arr = [5, 2, 8, 1, 3];
arr.sort((a, b) => a - b);
console.log("Sorted:", arr);
arr.push(10);
console.log("After push:", arr);
arr.pop();
console.log("After pop:", arr);
let squared = arr.map(num => num * num);
console.log("Squared:", squared);
let evens = arr.filter(num => num % 2 === 0);
console.log("Evens:", evens);
let sum = arr.reduce((acc, num) => acc + num, 0);
console.log("Sum:", sum);
```

ii) Describe the role of JSON in modern web APIs. How does it facilitate communication between clients and external applications?" (6)

c) Describe the Document Object Model (DOM) and its significance in web development. Create a webpage that contains a form for student registration with fields for name and email, and a submit button. When the submit button is clicked, the text of a paragraph below the form should change to 'Registration Successful!'. Draw the DOM tree structure and write the HTML, CSS, and JavaScript needed to achieve this.

4 a) Predict the order of execution of statements in the following JavaScript code. Also give reasons.

```
console.log('Start');
setTimeout(() => console.log('on your mark'),400);
setTimeout(() => console.log('set'),200);
console.log('go');
console.log('End');
```

b) i) Compare the transaction handling capabilities of SQL and NoSQL databases, and justify which one is better suited for high-volume transactions. (5)

ii) How does Node.js manage multiple concurrent requests without blocking, and what role does the event loop play in enabling non-blocking I/O operations?" (5)

c) i) How do synchronous and asynchronous programming differ in JavaScript, and what are the advantages and disadvantages of each approach when handling tasks like file I/O, API requests, and event handling?

(OR)

ii) How can you analyze the MVC architecture and its components to structure an ATM login page application using JS Technologies, ensuring efficient management of user authentication and interface.

/END/

FD/JU